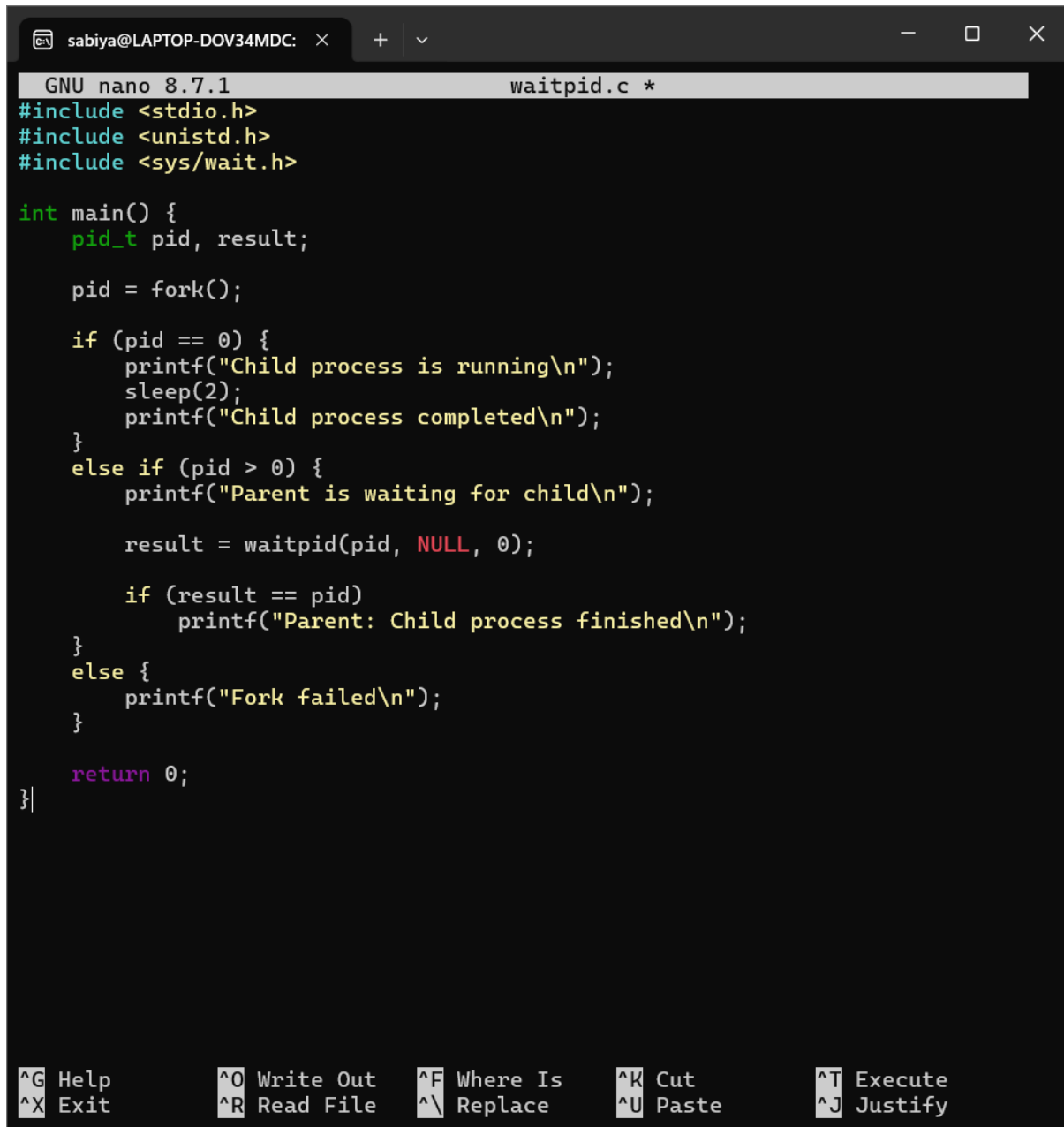


## 1. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

Code:



```
GNU nano 8.7.1 waitpid.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid, result;

    pid = fork();

    if (pid == 0) {
        printf("Child process is running\n");
        sleep(2);
        printf("Child process completed\n");
    }
    else if (pid > 0) {
        printf("Parent is waiting for child\n");

        result = waitpid(pid, NULL, 0);

        if (result == pid)
            printf("Parent: Child process finished\n");
    }
    else {
        printf("Fork failed\n");
    }

    return 0;
}
```

^G Help      ^O Write Out      ^F Where Is      ^K Cut      ^T Execute  
^X Exit      ^R Read File      ^\ Replace      ^U Paste      ^J Justify

Screenshot:

```
Parent process completed
sabiya@LAPTOP-DOV34MDC:~$ nano waitpid.c
sabiya@LAPTOP-DOV34MDC:~$ gcc waitpid.c -o waitpid
sabiya@LAPTOP-DOV34MDC:~$ ./waitpid
Parent is waiting for child
Child process is running
Child process completed
Parent: Child process finished
sabiya@LAPTOP-DOV34MDC:~$ |
```